

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:10-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

10-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-12(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

PHYSICS

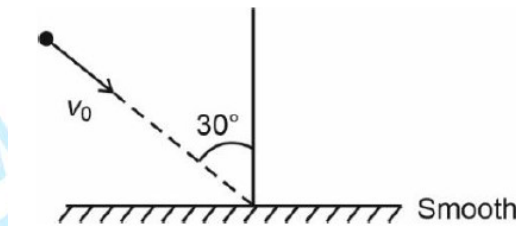
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A billiard ball of mass m moving with speed v_0 strikes a smooth floor at an angle of 30° with normal to floor. If ball rebounds at an angle of 60° with vertical, then coefficient of restitution is



- 1) $1/2$ 2) $1/3$ 3) $1/2\sqrt{3}$ 4) $1/3\sqrt{2}$
2. A plano-convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered at the curved surface. Now this lens has been used to form the image of an object. At what distance from this lens an object is placed in order to have a real image of the size of the object ?
- 1) 20 cm 2) 80 cm 3) 60 cm 4) 30 cm
3. **STATEMENT – 1:** Kinetic energy of a particle is conserved if it is acted upon by a conservative force only.
- STATEMENT – 2:** Work done by a conservative force in a closed path is zero.
- (A) Statement – 1 is True, Statement – 2 is True; Statement – 2 is a correct explanation for Statement – 1.
- (B) Statement – 1 is True, Statement – 2 is True; Statement – 2 is **NOT** a correct explanation for Statement – 1.
- (C) Statement – 1 is True, Statement – 2 is False.
- (D) Statement – 1 is False, Statement – 2 is True.

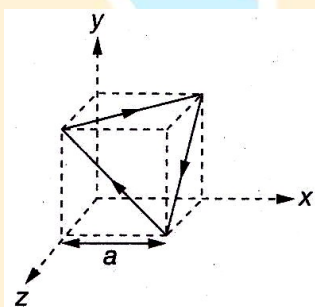
4. A thick-walled hollow sphere has outer radius R . It rolls down on an inclined plane without slipping and its speed at bottom is v_0 . Now the incline is waxed so that the friction becomes zero. The sphere is observed to slide down without rolling and the speed now is $(5v_0/4)$. The radius of gyration of the hollow sphere about the axis through center is $\frac{nR}{4}$. Then the value of n is

1)1 2)4 3)3 4)2

5. Electric potential V in volt in a region is given by $V = ax^2 + ay^2 + 2az^2$, where a is a constant. Work done by the field when a $2\mu\text{C}$ charge moves from point $(0,0,0.1\text{m})$ to origin is $5 \times 10^{-5} \text{ J}$. Find a ? (in V/m^2)

1)1250 2)1520 3)1750 4)1500

6. Calculate the magnetic moment associated with the loop carrying current I_0 as shown in figure is



1) $\frac{3\sqrt{3}}{2} I_0 a^2$ 2) $\frac{2\sqrt{3}}{2} I_0 a^2$ 3) $\frac{2}{3} I_0 a^2$ 4) $\frac{\sqrt{3}}{2} I_0 a^2$

7. What mass (approximately) of coal with calorific value of 30 kJ/g is thermally equivalent to the heat liberated during the formation of one gram of He^4 from deuterium H^2 ?

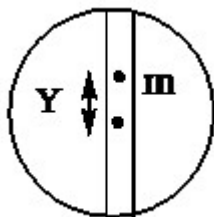
$$m(\text{H}^2) = 2.01410 \text{ amu}, m(\text{He}^4) = 4.002603 \text{ amu}$$

1) $1 \times 10^4 \text{ kg}$ 2) $2 \times 10^4 \text{ kg}$ 3) $3 \times 10^4 \text{ kg}$ 4) $4 \times 10^4 \text{ kg}$

8. A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 W of power. The amplitude of the electric field associated with the electromagnetic radiation at a distance of 10 m from the lamp will be

1)1.34 V/m 2)2.68 V/m 3)5.36 V/m 4)9.37 V/m

9. Suppose a vertical tunnel is dug along the diameter of the earth assumed to be a sphere of uniform mass having density ρ . If a body of mass m is thrown in this tunnel, its acceleration at a distance y from a centre is given by

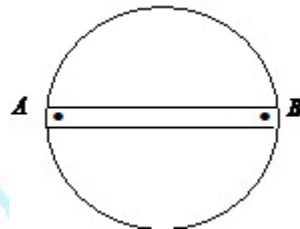


- 1) $\frac{4\pi}{3} g \rho y$ 2) $\frac{3}{4} \pi G \rho y$ 3) $\frac{4}{3} \pi \rho y$ 4) $\frac{4}{3} \pi G \rho y$
10. A vessel contains liquid of density ρ up to height h . Vessel is given horizontal acceleration a_0 such that free surface of liquid makes an angle of 30° with horizontal. Value of a_0 is

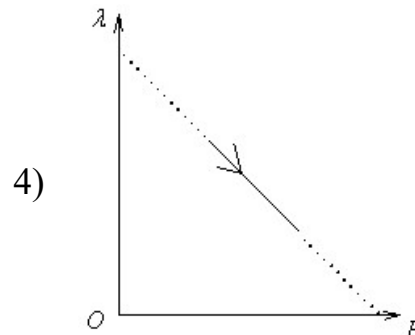
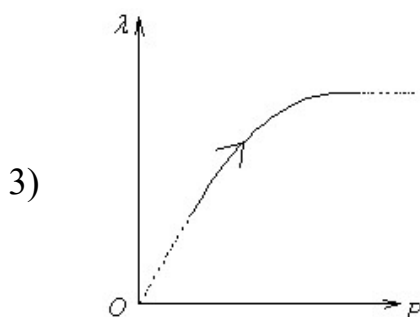
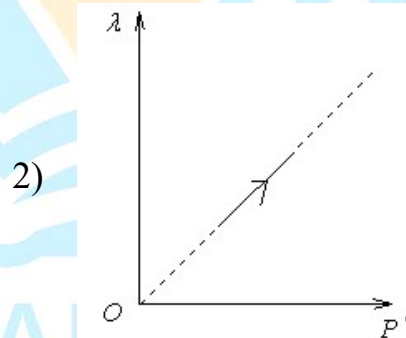
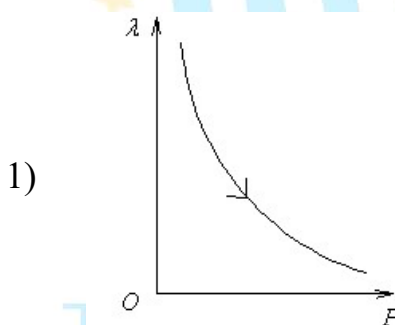


- 1) $g/2$ 2) $g/3$ 3) $g/\sqrt{3}$ 4) $\sqrt{3}g/2$
11. A projectile is projected with velocity 20 m/s at an angle of 53° with horizontal. Speed of the projectile when its velocity is perpendicular to its initial velocity, is
- 1) 16 m/s 2) 12 m/s 3) 15 m/s 4) 18 m/s
12. A wire of length $l = 6 \pm 0.06 \text{ cm}$ and radius of cross-section $r = 0.5 \pm 0.005 \text{ cm}$ and mass $m = 0.3 \pm 0.003 \text{ gm}$. Maximum percentage error in density is
- 1) 4 2) 2 3) 1 4) 6.8
13. In Young's double slit experiment using monochromatic light, the fringe patterns shifts by a certain distance on the screen when a mica sheet of refractive index 1.6 and thickness 1.964 micron is introduced in the path of one of the interfering waves. The mica sheet is then removed and the distance between the plane of slits and the screen is doubled. It is found that the distance between successive maxima now is the same as the observed fringe shift upon the introduction of the mica sheet. The wavelength of light is
- 1) $5762 \text{ \AA}$ 2) $5825 \text{ \AA}$ 3) $5892 \text{ \AA}$ 4) $6500 \text{ \AA}$

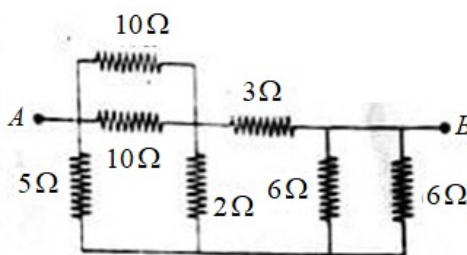
14. Suppose a narrow, and straight tunnel passes through the center of earth as shown in the figure. Two small balls A and B are released simultaneously at the two ends of the tunnel. The balls collide with each other and after collision one ball comes to rest and the other ball just manages to escape the earth's gravitational field. If the coefficient of restitution between the balls is e , then select the correct option.



- 1) $e = \frac{\sqrt{3}}{2}$ 2) $e = \frac{1}{\sqrt{2}}$
- 3) $\frac{1}{\sqrt{2}} < e < \frac{\sqrt{3}}{2}$ 4) information is insufficient to predict
15. An ideal gas undergoes an isothermal process. The pressure (P) of the gas is plotted against the mean free path λ of the molecules. Select the correct graph.



16. If two soap bubbles of different radii are in communication with each other, then
- 1) air flows from the larger bubble into the smaller one until their sizes become equal
 - 2) the sizes of the bubbles remain unchanged
 - 3) air flows from the smaller bubbles into the larger one and the larger bubble grows at the expense of the smaller one
 - 4) air flows from the larger into the smaller bubble until their radii interchange
17. A copper wire and a steel wire of the same diameter and length are connected end to end. A force is applied which stretches their combined length by 1 cm. Then the two wires have
- 1) the same stress and strain
 - 2) the same stress but different strains
 - 3) the same strain but different stresses
 - 4) different stresses and strains
18. An energy of 24.6 eV is required to remove one of the electrons from a helium atom. The energy (in eV) required to remove both the electrons from a neutral helium atom is
- 1) 38.2
 - 2) 49.2
 - 3) 51.8
 - 4) 79.0
19. If 200 MeV energy is released per fission of ${}_{92}\text{U}^{235}$, how many fissions must occur per second to produce a power of 1 mW?
- 1) 6.25×10^7
 - 2) 12.25×10^7
 - 3) 25×10^7
 - 4) 3.125×10^7
20. Find the equivalent resistance between points A and B (in Ω)_



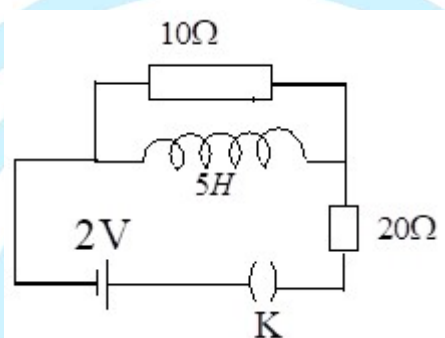
- 1) 4
- 2) 6
- 3) 11.2
- 4) 10.4

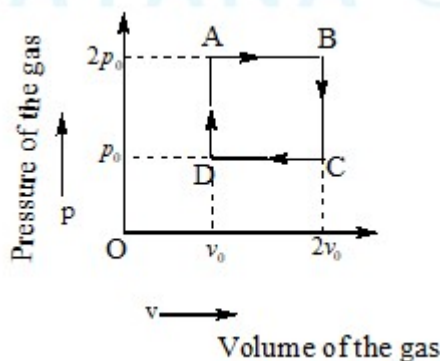
SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

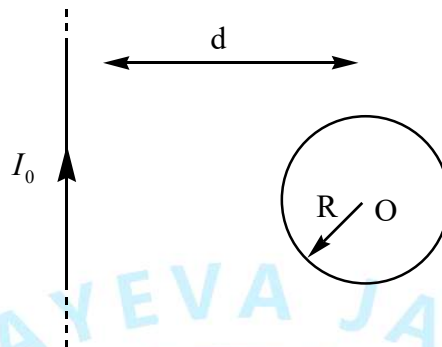
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

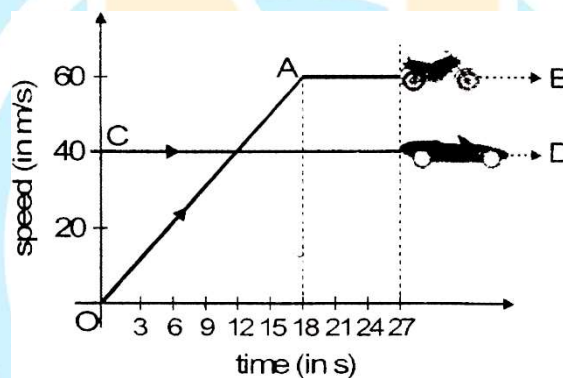
21. A uniform string of length 9m and mass 4.5 kg is fixed at one end and hangs freely under its own weight. A wave pulse is generated at top end which runs down to other end. At the same moment a stone is released from rest and falls freely from the top of string. How far from the top does stone pass the wave ?
22. Two resistors of 10Ω and 20Ω and an ideal inductor of 5 H are connected to a 2 V battery as shown in the below figure. The key is plugged in at $t = 0$ the value of the current in the 10Ω resistor at steady state is
- 
23. A person can throw a ball upto a maximum range of 100m. The maximum height of this projectile is
24. The p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monoatomic gas. The amount of heat extracted from the source in a single cycle is $\left(\frac{x}{2}\right)p_0v_0$. Find the value of x.



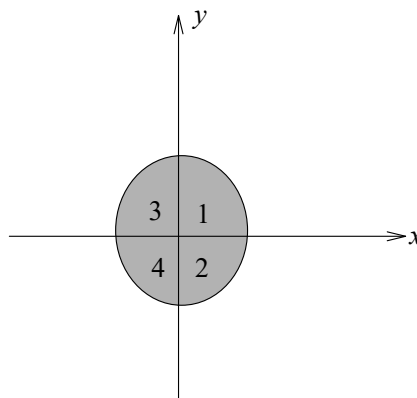
25. Current, I_0 flows in long straight conductor as shown. If magnetic field at center of circular loop in the same plane is zero, then current in the circular loop is $\frac{|n|I_0R}{\pi d}$, then the value of $|n|$ is



26. At the instant a motor bike starts from rest in a given direction, a car overtakes the motor bike, both moving in the same direction. The speed time graphs for motor bike and car are represented by OAB and CD respectively. Then at $t = 18$ s find the distance between then motor bike and car.

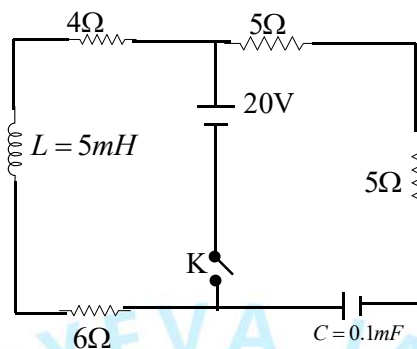


27. A solid disc of uniform thickness has density that varies by quadrants as shown, with number indicating relative densities. If x-y axes are as indicated with centre of disc at origin, then the equation of straight line drawn through origin and centre of mass of the disc is $y = nx$. Find the value of n

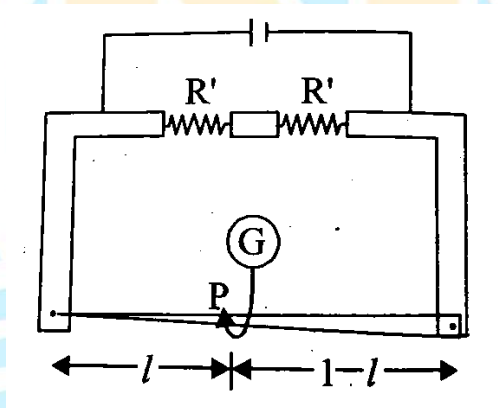


28. In the circuit shown, the key (K) is closed at $t=0$, the current through the key at the

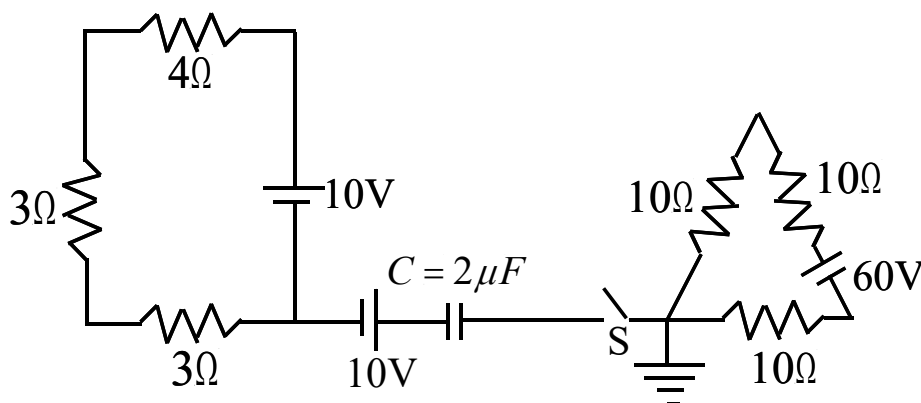
$t=10^{-3} \ln 2$ s is $\frac{x}{2}$ ampere. Find the value of x



29. In a meter bridge, the wire of length 1m has a non – uniform cross-section such that, the variation $\frac{dR}{dl}$ of its resistance R with length l is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. The length of AP is $\frac{x}{4}$ m. Find the value of x



30. In the circuit shown in figure the switch S is closed at $t=0$, find the energy stored on the capacitor at steady state.



SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Match the Column-I with Column-II

Species	Shape
A) $S_2O_3^{2-}$	1) pyramidal
B) ClO_3^-	2) linear
C) C_3O_2	3) squareplanar
D) $Ni(CO)_4$	4) tetrahedral

1) A-3, B-1, C-2, D-4

2) A-4, B-1, C-2, D-3

3) A-2, B-1, C-3, D-4

4) A-4, B-1, C-2, D-4

32. The Solubility of Calcium fluoride in water is $7.8 \times 10^{-4} \text{ g/L}$. The value of $\log_{10} K_{sp}$ of calcium fluoride is

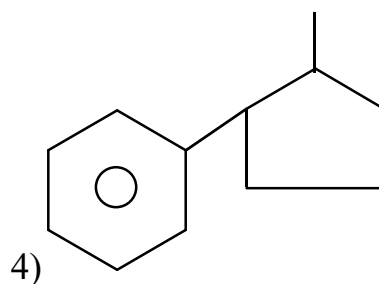
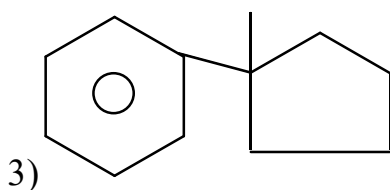
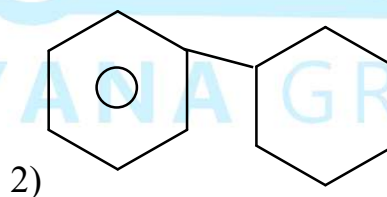
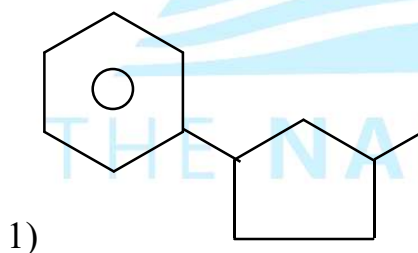
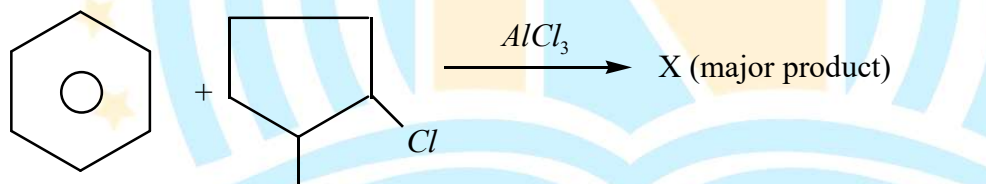
1) 4×10^{-15}

2) -14.4

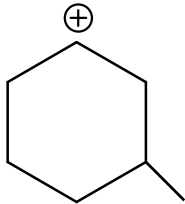
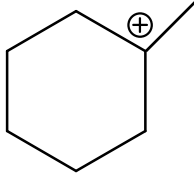
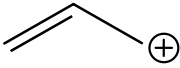
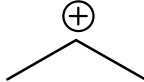
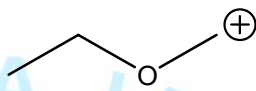
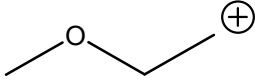
3) 14.4

4) -4×10^{15}

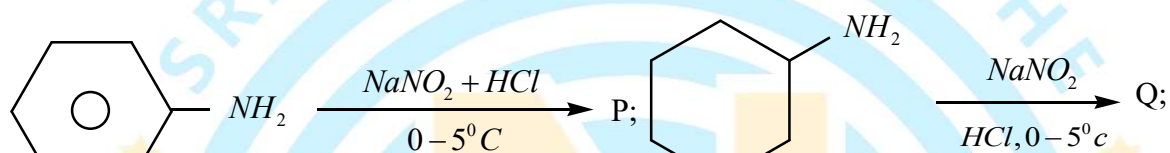
33.



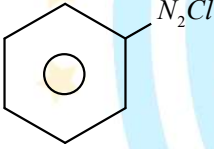
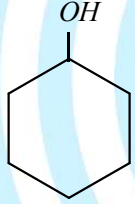
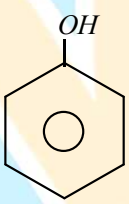
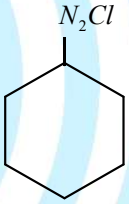
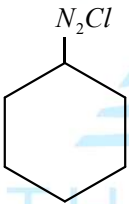
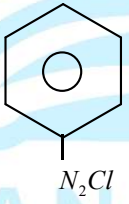
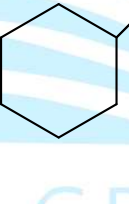
34. In which of following pairs "A" is less stable than "B"?

- 1)  (A)  (B)
- 2) $Ph-CH_2^+$ (A) $C_6H_5^+$ (B)
- 3)  (A)  (B)
- 4)  (A)  (B)

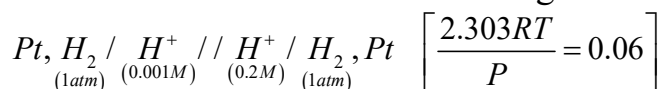
35.



P and Q are

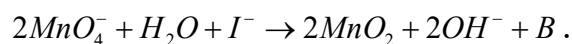
- 1)  
- 2)  
- 3)  
- 4)  

36. Electromotive force of the following cell at 298K



- 1) 120mv 2) 0.12V 3) 138mv 4) 0.138mv

37. $2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{+2} + 8H_2O + A$,



A & B respectively are

- 1) IO_3^-, I_2 2) I_2, IO_3^- 3) I_2, I_2 4) IO_3^-, IO_3^-

38. The number of geometric isomers possible for square planar complex



- 1) 6 2) 8 3) 3 4) 12

39. Correct statements among the following are

A) At isoelectric point amino acids having least solubility

B) All natural amino acids are optically active

C) Globular proteins have coiled (spherical) like structure and are water soluble

D) Fibrous proteins have sheet like (run in parallel) structure and are water soluble

- 1) A, B only 2) B, C only 3) C, D only 4) A, C only

40. The radius of $La^{+3} (z = 57)$ is 108 pm . The radius of $Lu^{+3} (z = 71)$ will be closest to

- 1) 85 pm 2) 108 pm 3) 180 pm 4) 160 pm

41. Consider following statements about complexes

A) $FeSO_4 \cdot (NH_4)_2 SO_4 \cdot 6H_2O$ and $FeSO_4 \cdot 4KCN$ are complex compounds

B) All Zn^{+2} complexes (C.N=4 and 6) are diamagnetic

C) In $K_4[Fe(CN)_6]$, iron undergoes d^2sp^3 hybridisation

D) $[Cu(NH_3)_4]SO_4$ is square planar complex

Select the correct statement(s) from the above?

- 1) A, B, C only 2) B, C, D only 3) A, C, D only 4) A, B, C, D

42. Enthalpy change for the conversion of $\frac{1}{2}Cl_2(g)$ to $Cl^{-1}(aq)$ is _____ KJ/mole

(Given that $\Delta_{dis}H(Cl_2) = 240 \text{ KJ/mol}$, $\Delta_{eg}H(Cl_g^{-}) = -350 \text{ KJ/mol}$ and

$$\Delta_{hyd}H(Cl_g^{-}) = -380 \text{ KJ/mol}$$

- 1) -610 2) 610 3) -490 4) 490

43. The species given that doesn't undergoes disproportionation reaction is

- 1) Cl_2 2) MnO_4^{-} 3) BrO_2^{-} 4) H_2O_2

44. Match List-I & List-II

List-I (Atomic Number)	List-II (Block of periodic table)
A) 56	P) d-block
B) 49	Q) f-block
C) 79	R) p-block
D) 64	S) s-block

1) A-R, B-P, C-S, D-Q

2) A-S, B-P, C-Q, D-P

3) A-S, B-R, C-P, D-Q

4) A-S, B-R, C-Q, D-P

45. The chalcogen with highest negative electron gain enthalpy is

1) O

2) Se

3) S

4) Te

46. The nitrogen of following compound doesn't converted into ammonium sulphate, in estimation of Nitrogen by Kjeldhal's method.

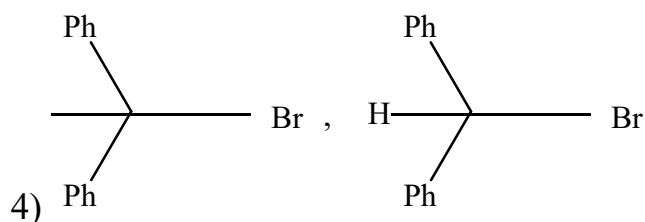
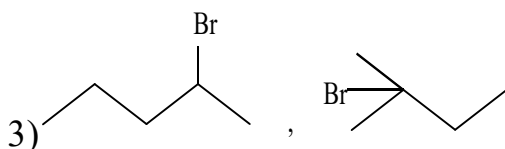
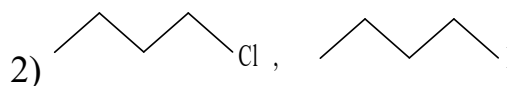
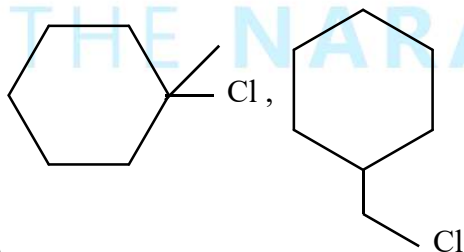
1) propanamine

2) urea

3) Aniline

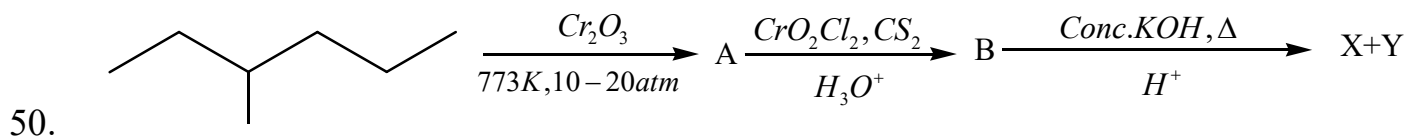
4) Nitro benzene

47. Identify the INCORRECT combination

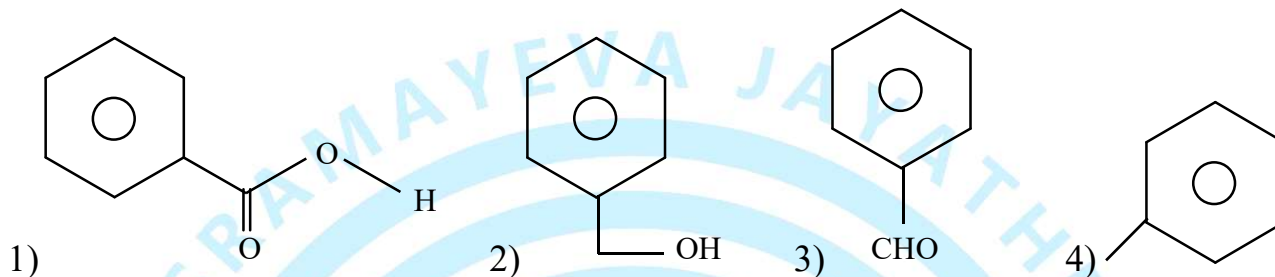
1) Oxidising power - $F_2 > Cl_2 > Br_2 > I_2$ 2) Bond energy - $Cl_2 > Br_2 > F_2 > I_2$ 3) S.R.P - $F_2 < Cl_2 < Br_2 < I_2$ 4) Water solubility at $25^\circ C$ - $Ne < Ar < Kr < Xe$ 48. In which of following pairs of halogen compounds first one undergoes SN^2 reaction faster.

49. Molal depression constant for a solvent 5.0 kg mol^{-1} . The depression in freezing point of solvent for 0.4 mole kg^{-1} solution of Na_2SO_4 is (undergoes 100% ionisation).

- 1) 3 2) 4 3) 2 4) 6



X gives "B" when treated with MnO_2 . Then Y is



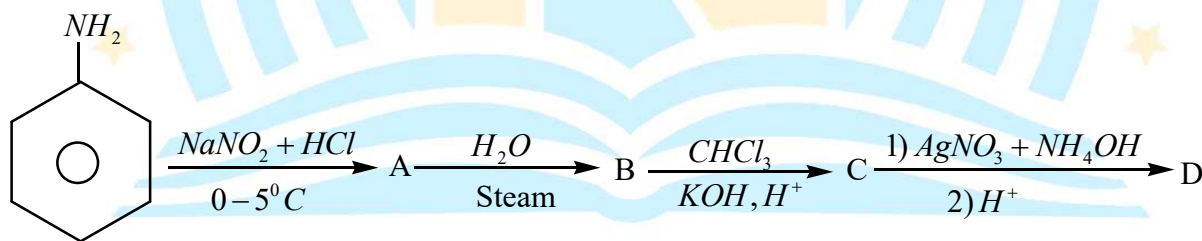
SECTION-II (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

51. The de-Broglie wave length of electron present in second Bohr orbit of H-atom is πA^0 (Round off to nearest integer)

52.

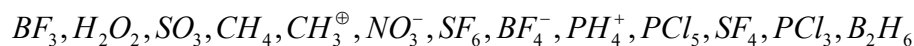


Molecular weight of D is g/mol .

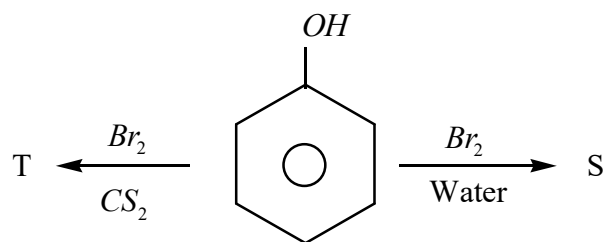
(Atomic weight of H=1, C=12, N=14, O=16 g/mol).

53. At 25°C , the pH of $1.0 \times 10^{-8} \text{ M}$ HCl aqueous solution is _____
(Round off to nearest integer).

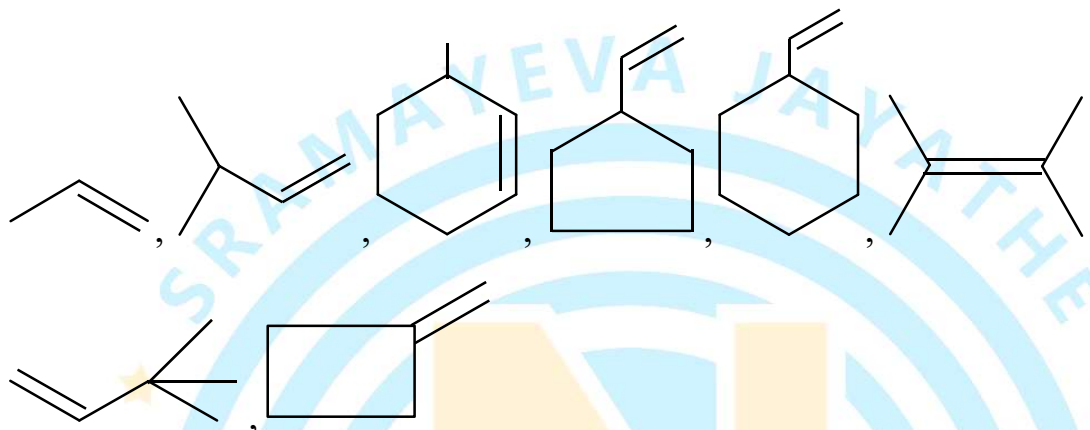
54. The number of molecules or ions from the following having non-planar structure is _____



55.



The difference between molecular weights of "S" and "T" is _____ (g)
(Atomic weight H=1, C=12, O=16, Br=80 g/mol).

56. Number of alkenes when treated with HCl gives rearranged product

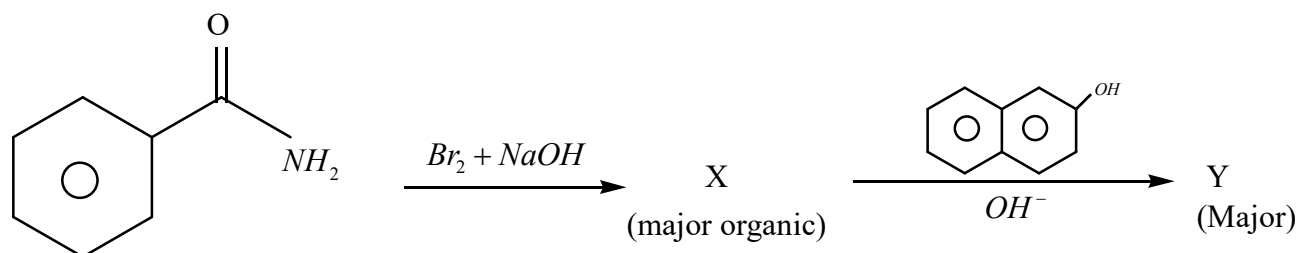
57. How many of following reduces ammoniacal silver nitrate solution?

Glucose, Fructose, Galactose, Sucrose, Maltose, Lactose, Ribose, De-oxy ribose, Starch, Cellulose.

58. The d-electronic configuration of $[Ru(en)_3]Cl_2$ is $t_{2g}^x e_g^y$ and that of $[Fe(H_2O)_6]Cl_2$ is $t_{2g}^z e_g^p$. Then $[12(x+z+p-y)]$ is

59. Rate constant of a reaction is 2.303 sec^{-1} . Calculate $t_{75\%} + t_{99\%}$ value of same reaction is _____ $\times 10^{-1} \text{ sec}$

60.



Degree of unsaturation in "Y" is _____

MATHEMATICS**MAX.MARKS: 100****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. If $\alpha = \cos\left(\frac{8\pi}{11}\right) + i\sin\left(\frac{8\pi}{11}\right)$, then $\operatorname{Re}(\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5)$ is equal to
- 1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) 0 4) 11
62. The mean of two samples of sizes 200 and 300 were found to be 25, 10 respectively. Their standard deviations were 3 and 4 respectively. The variance of combined sample of size 500 is
- 1) 6.72 2) 67.2 3) 672 4) 0.672
63. The sum of the series $\frac{7}{2^3} + \frac{19}{6^3} + \frac{37}{12^3} + \frac{61}{20^3} + \dots - \infty$
- 1) 1 2) 2 3) 3 4) 4
64. If x_1, x_2 and x_3 are the positive roots of the equation $x^3 - 6x^2 + 3px - 2p = 0, p \in \mathbb{R} - \{0\}$ then the value of $\sin^{-1}\left(\frac{1}{x_1} + \frac{1}{x_2}\right) + \cos^{-1}\left(\frac{1}{x_2} + \frac{1}{x_3}\right) - \tan^{-1}\left(\frac{1}{x_3} + \frac{1}{x_1}\right)$ is equal to
- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{3\pi}{4}$ 4) π
65. Let $f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ for any real number x , and let g be the inverse function of f . Then the value of $g'\left(-\frac{7}{6}\right)$ is
- 1) 5 2) $\frac{1}{5}$ 3) 3 4) $\frac{1}{3}$
66. Let the function $g : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(t) = \frac{\pi}{2} - 2 \cot^{-1}(3^{-t})$.
- P : g is an odd function
- Q : g is strictly increasing in $(-\infty, \infty)$.
- 1) P is true ; Q is true 2) P is true ; Q is false
- 3) P is false ; Q is true 4) P is false ; Q is false

67. $y = f(x)$ is a solution of $\frac{dy}{dx} \left(\frac{1 + \cos x}{y} \right) = -\sin x$ and $f\left(\frac{\pi}{2}\right) = 1$ then $f(0)$ is

- 1) 2 2) 1 3) 3 4) 4

68. Let a straight line passing through $P(1, 4)$ with negative slope cuts the coordinate axes at A, B then the area of the triangle OAB when $OA + OB$ is minimum is _____

- 1) 9 2) 18 3) 4 4) 14

69. If $\bar{a}, \bar{b}$ & $\bar{c}$ are such that $(\bar{a} \times \bar{b}) \cdot \bar{c} = 1$, $\bar{c} = \lambda \bar{a} \times \bar{b}$, $|\bar{a}| = \sqrt{2}$, $|\bar{b}| = \sqrt{3}$ & $|\bar{c}| = \frac{1}{\sqrt{3}}$, then the angle between $\bar{a}$ & $\bar{b}$ is

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

A) Assertion: There exists two points on the line $\frac{x-1}{1} = \frac{y}{-1} = \frac{z+2}{2}$ which are at a distance of 2 units from the point $(1, 2, -4)$

(R) Reason : Perpendicular distance of point $(1, 2, -4)$ from the line $\frac{x-1}{1} = \frac{y}{-1} = \frac{z+2}{2}$ is 1 unit,

- 1) A, R are true and R is correct explanation of A
2) A, R are true but R is not correct explanation of A
3) A is true and R is false
4) R is true and A is false

71. If ABCD is a square of unit side, 4-circles of unit radius are described with centres at A, B, C, D then area common to 4 – circles is

- 1) $1 - \frac{\pi}{4} + \sqrt{3}$ 2) $1 + \frac{\pi}{4} - \frac{\sqrt{3}}{2}$ 3) $1 + \frac{\pi}{3} - \sqrt{3}$ 4) $1 - \frac{\pi}{3} + \sqrt{3}$

72. Let $ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h = \begin{vmatrix} x+1 & x^2+2 & x^2+x \\ x^2+x & x+1 & x^2+1 \\ x^2+2 & x^2+x & x+1 \end{vmatrix}$ then

- 1) $g = 3$ and $h = -5$ 2) $g = -3$ and $h = -5$
3) $g = 3$ and $h = 9$ 4) $g = -2$ and $h = 5$

73. If the equation $\sin^4 x - (k+2)\sin^2 x - (k+3) = 0$ has a solution then k must lie in the interval:

- 1) $(-4, -3)$ 2) $(-2, 0)$ 3) $(-2, 2)$ 4) $[-3, -2]$

74. If 'a' be the digit at unit's place in $11^{2012} + 23^{2012} - 3^{2012}$, then $\int_{a-1}^a \frac{dx}{\sqrt{1-x^2} - x + \frac{1}{x}} =$

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{4}$

75.

	Column-I		Column-II
A	Area bounded by $y = x $ and $y=2$ is	p	4
B	Area bounded by $\frac{ x }{a} + \frac{ y }{b} = 1$, when $a, b > 0$ is	q	$\frac{(\pi-2)ab}{4}$
C	Area between the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord $\frac{x}{a} + \frac{y}{b} = 1$ ($a, b > 0$) is	r	1
D	Area bounded by $y = [x]$, the x-axis and $x = 1, x = 2$ is [.] denotes greatest integer function	s	2ab

- 1) A-S, B-P, C-Q, D-R 2) A-P, B-S, C-Q, D-R
3) A-S, B-P, C-R, D-Q 4) A-Q, B-S, C-R, D-P

76. If the probability that the random variable X takes values x is given by $P(X = x) = \frac{K}{3^x}$

$x = 0, 1, 2, 3, \dots$ where K is constant then $P(x \geq 2) =$

- 1) $\frac{1}{3}$ 2) $\frac{1}{27}$ 3) $\frac{1}{18}$ 4) $\frac{1}{9}$

77. Let $S = \{1, 2, 3, \dots, 50\}$. The number of non empty subsets A of S such that the product of elements of A is even

- 1) $2^{25}(2^{25}-1)$ 2) $2^{25}-1$ 3) $2^{25}+1$ 4) $2^{50}-1$

78. The number of ordered pairs (m, n) where $m, n \in \{1, 2, 3, \dots, 50\}$, such that $6^m + 9^n$ is a multiple of 5 is

- 1) 1250 2) 2500 3) 625 4) 500

79. A set S contains 7 elements. A non-empty subset A of S and an element x of S are chosen at random. Then the probability that $x \in A$ is:

- 1) $\frac{1}{2}$ 2) $\frac{64}{127}$ 3) $\frac{63}{128}$ 4) $\frac{31}{128}$

80. $\lim_{x \rightarrow 0} \frac{\tan(1 - \{x\}) \sin \{x\} \cos \{x\}}{\{x\}(1 - \{x\})}$

(where $\{x\}$ denotes fractional part and $[x]$ denotes greatest integer)

- 1) 1 2) $\tan 1$ 3) $\cos 1$ 4) does not exist

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81. Consider the set of eight vectors $V = \{a\hat{i} + b\hat{j} + c\hat{k} ; a, b, c \in \{-1, 1\}\}$. Three non-coplanar vectors can be chosen from V in 2^p ways. Then p is

82. Let $0 \leq a, b, c, d \leq \pi$, where b and c are not complementary, such that $2 \cos a + 6 \cos b + 7 \cos c + 9 \cos d = 0$ and such that

$2 \sin a - 6 \sin b + 7 \sin c - 9 \sin d = 0$, then the value of $3 \frac{\cos(a+d)}{\cos(b+c)}$ is _____

83. Given $(1+x+x^2)^8 = \sum_{k=0}^{16} a_k x^k$ then the value of $\begin{pmatrix} a_3 & a_7 & a_{13} \\ a_4 & a_6 & a_{12} \\ a_5 & a_4 & a_{11} \end{pmatrix} = \dots\dots\dots$

84. If $S_n = \sum_{r=1}^n \tan^{-1} \left(\frac{2(2r-1)}{4+r^2(r^2-2r+1)} \right)$ then find the value of $\lim_{n \rightarrow \infty} \sum_{k=2}^n (\cot(S_{k-1}) - \cot(S_k))$.

85. If $I = \int (x^2 + 1)((x+1)e^x)^2 dx = A(f(x))^2 + C$, where C is constant of integration and $f(-1) = \frac{2}{e}$, then $2A + f(0)$ is
86. Let three matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$ then $T_r(A) + T_r\left(\frac{ABC}{2}\right) + T_r\left(\frac{A(BC)}{4}\right) + \dots =$ (where T_r denotes trace of matrix)
87. Mr.A has two fair cubic dice one with faces numbered from 2 to 7 and the second with faces numbered from 4 to 9. Twice, he randomly picks one of the dice (selection of each dice is equally likely) and rolls. If it is known that the sum of the resulting two rolls is 10 then the probability he rolled the same dice twice is $\frac{m}{n} (m, n \in \mathbb{N})$, (G.C.D. (m,n)=1), find the least value of $(n-m)$.
88. A is targeting to B. B and C are targeting to A. The probability of hitting the targets by A, B, C are $\frac{2}{3}, \frac{1}{2}$ and $\frac{1}{3}$ respectively. If A is hit, then the probability that B hits the target and C does not is P then value of 6P is _____
89. 6 letters $L_1, L_2, \dots, L_6$ be inserted into 6 addressed envelopes $E_1, E_2, \dots, E_6$ one letter each into one envelope such that no letter goes into its corresponding envelop. If the number of ways in which letter L_2 is placed in envelop E_3 is N then value of $\left[\frac{N}{13}\right]$ is (where $[.]$ is GIF).
90. If $x, y \in \mathbb{R}$ and $\log_4(x+2y) + \log_4(x+2y) = 1$. Then, the minimum value of $(|x| - |y|)^2$ is